

Multiple Choice (10 marks)

1. Use a number line to find the sum of $-9 + (-8)$.
 - (a) -17
 - (b) 1
 - (c) -1
 - (d) 17
2. Jonathan obtained a score of 80 on a statistics exam, placing him at the 90 th percentile. Suppose five points are added to everyone's score. Jonathan's new score will be at the
 - (a) 80 th percentile.
 - (b) 85 th percentile.
 - (c) 90 th percentile.
 - (d) 95 th percentile.
3. Find the volume of the solid generated when the region bounded by the y-axis, $y = e^x$, and $y = 2$ is rotated around the y-axis.
 - (a) 0.296
 - (b) 0.592
 - (c) 2.427
 - (d) 3.998
4. How many positive cubes divide $3! \cdot 5! \cdot 7!$?
 - (a) 6
 - (b) 4
 - (c) 3
 - (d) 1
5. Two whole numbers have a least common multiple of 60. Each number is less than or equal to 12. The greatest common factor of the two numbers is 2. What are the two numbers?
 - (a) 6 and 10
 - (b) 5 and 12
 - (c) 10 and 12
 - (d) 12 and 15

True / False (10 marks)

Answer True or False

6. True or False: The upper critical value of z for a 94% confidence interval is 1.88.
7. True or False: The difference in elevation between Talon Bluff and Salt Flats is 675 feet.
8. True or False: -5 is greater than -3 .
9. True or False: The solution to the equation $3/2 * w = 66$ is $w = 33$.
10. True or False: The expression simplifies to $x^{11} + 10x^9 + 7x^8 + 6x^7 + 3x^3 + 5x + 8$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of residuals in a regression line, particularly focusing on why the mean of the residuals is always zero and its implications for the model's accuracy.
12. In the first quadrant, analyze the rose $r = \sin 3$ to determine where it has a vertical tangent. Explain your reasoning and the significance of this point.

End of Paper